switch Conditional Branch

ELEC1006: ENGINEERING COMPUTING

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Conditional Branches

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- Defined as a condition that causes the program to change the path of execution based on the value of an expression.
- Forms of conditional branches:
  - *if* statement.
  - *switch* statement. ←

switch Statement

- `switch` statement is a natural choice for menu-driven program:
  - display the menu
  - then, get the user's menu selection
  - use user input as `expression` in `switch` statement
  - use menu choices as *`expr`* in `case` statements
- `if-else` can be used in place of `switch`.

switch Statement Format

```
switch (expression) //integer
{
  case exp1: statement1;
  case exp2: statement2;
  ...
  case expn: statementn;
  default:   statementn+1;
}
```

break Statement

- Used to exit a `switch` statement
- If it is left out, the program "falls through" the remaining statements in the `switch` statement
- Example:

```
switch (n) {
   case ’a':
      cout << ”a has been given.\n";
      break;
   case ’b':
      cout << ”b has been given.\n";
      break;
   default:
      cout << "unknown character!\n";
      break;
}
```

Alternative Program

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- Use `if - else if - else` statement as an alternative to `switch`:
- Example:

```
if (n == ‘a’)
     cout << ”a has been given.\n";
else if (n == ’b’)
     cout << ”b has been given.\n";
else
     cout << "unknown character!\n";
}
```

More info

[1] cplusplus.com: Statements and flow control
https://cplusplus.com/doc/tutorial/control/

[2] learncpp.com: 5.3 – Switch statements
https://www.learncpp.com/cpp-tutorial/53-switch-statements/